

ADVANCED AI MEDICAL INTELLIGENCE PLATFORM

Technical Evaluation Project Assignment Report | SN Matrix Software Pvt. Ltd.

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1. Executive Summary

This project presents an end-to-end clinical AI intelligence system built to assist medical practitioners in automated diagnostic screening of chest radiographs (X-Rays). The platform unifies PyTorch Deep Learning pathology classification, Grad-CAM visual Explainable AI (XAI), Large Language Model (LLM) radiology report generation, FastAPI REST web services, SQLAlchemy database tracking, and a glassmorphic Web Application UI.

2. System Architecture & Technology Stack

Layer	Technology Stack	Key Responsibilities
Deep Learning	PyTorch / Torchvision (DenseNet121)	Chest X-Ray pathology multi-class prediction (Normal, Pneumonia, COVID-19)
Explainable AI	Grad-CAM / OpenCV / NumPy	Convolutional gradient heatmap activation map generation & alpha overlay
LLM Reporter	Google Gemini 1.5 Flash / Clinical Logic	Structured medical report generation with diagnostic findings & recommendations
REST API	FastAPI / Uvicorn / Pydantic	High-performance async API endpoints with OpenAPI interactive documentation
Database	SQLAlchemy / SQLite	Persistent audit logging of patient metadata, predictions, heatmaps, and reports
Web Application	Vanilla Modern HTML5 / CSS3 / ES6 JS	Glassmorphic UI, side-by-side interactive opacity slider, and live history dashboard

3. Deep Learning Model & Explainable AI Engine

Model Design: The system utilizes a DenseNet121 backbone fine-tuned for thoracic medical imaging. DenseNet's dense connectivity promotes feature reuse, reducing vanishing gradients and making it exceptionally suited for subtle radiological pattern extraction.

Grad-CAM Formulation: Feature attributions are computed at the final dense block layer (features.denseblock4.denselayer16). Gradients of the target class score y^c with respect to feature activation maps A^k are global-average pooled to obtain neuron weights α_k^c :

$$\alpha_k^c = (1/Z) \sum_i \sum_j (\partial y^c / \partial A_{\{i,j\}}^k)$$

The resulting coarse heatmap is generated by combining weighted activations passed through a ReLU nonlinearity.

4. LLM Medical Report Generation Pipeline

The platform incorporates Google Gemini 1.5 Flash API for automated radiology report generation. The LLM receives structured inputs combining patient demographics, model confidence metrics, class probability distributions, and Grad-CAM spatial coverage indicators. If an API key is unconfigured, the system automatically falls back to an offline Medical Logic Generation Engine ensuring 100% operational uptime.

5. REST API Specifications & OpenAPI Endpoints

Endpoint	Method	Description
/api/v1/predict	POST	Upload medical image -> Returns prediction, Grad-CAM heatmap, DB record, and LLM report
/api/v1/gradcam	POST	Generate custom Grad-CAM heatmap visualization with adjustable alpha overlay
/api/v1/report/pdf/{scan_id}	GET	Generate and download printable PDF clinical radiology report
/api/v1/history	GET	Retrieve searchable/filterable database history of patient diagnostic records
/api/v1/analytics	GET	Fetch aggregate platform statistics, class breakdowns, and model confidence metrics
/api/v1/health	GET	System status, model readiness, and memory diagnostic checks

6. Database Schema Design

The relational database schema is modeled in SQLAlchemy and stored in SQLite/PostgreSQL. The scan_records table tracks scan_id, patient_id, patient demographics, image file references, probability vectors, spatial metrics, report text, risk stratification, and timestamps.

7. Docker & Local Deployment Setup

Containerized Launch: docker-compose up --build

Local Python Execution: python train_and_setup.py followed by python run.py

Interactive Web UI: Accessible at http://localhost:8000/

Swagger API Docs: Accessible at http://localhost:8000/docs
